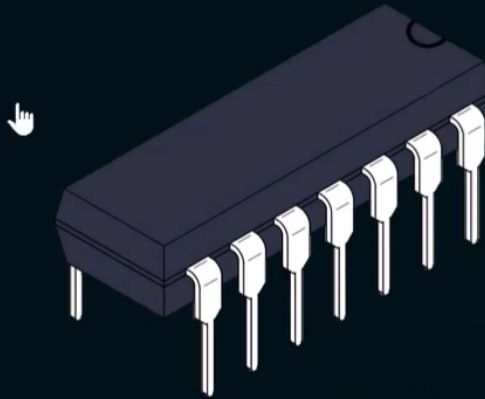


Evolution of Microprocessor



Jean Hoerni

FAIRCHILD
SEMICONDUCTOR®



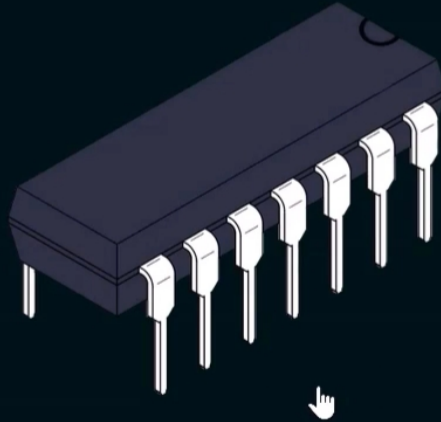
Robert Noyce



Evolution of Microprocessor



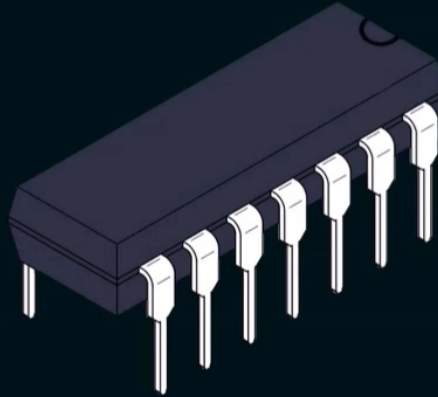
Gordon Earle Moore
(3 January 1929 – 24 March 2023)



Evolution of Microprocessor



Gordon Earle Moore
(3 January 1929 – 24 March 2023)

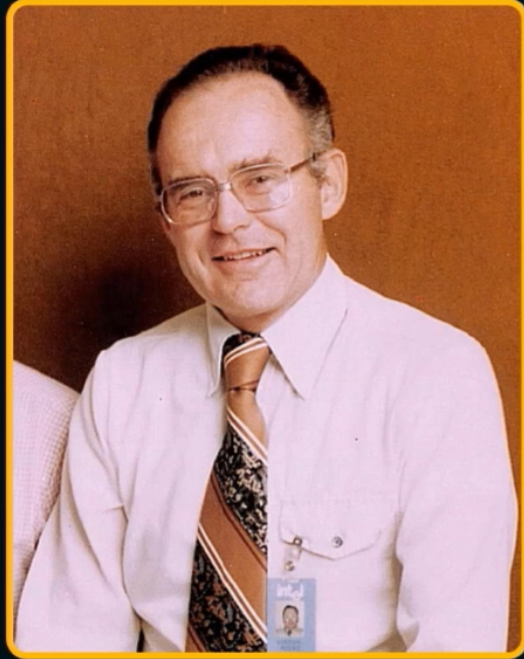


Moore's Law:

*"The number of transistors
in an integrated circuit will
double about every two
years."*



Evolution of Microprocessor



Gordon Earle Moore
(3 January 1929 – 24 March 2023)



Robert Noyce



IBM System/360



Evolution of Microprocessor



Gordon Earle Moore
(3 January 1929 – 24 March 2023)



DRAMs
EPROMs



Robert Noyce



Evolution of Microprocessor



Gordon Earle Moore
(3 January 1929 – 24 March 2023)



Robert Noyce



Evolution of Microprocessor

⇒ **1971**: Intel 4000 family

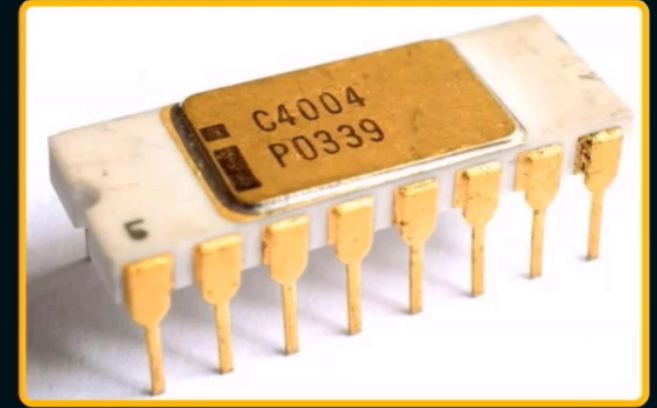
4001: 256-Byte ROM

4002: 40-Byte RAM

4003: 10-bit I/O Shift Register

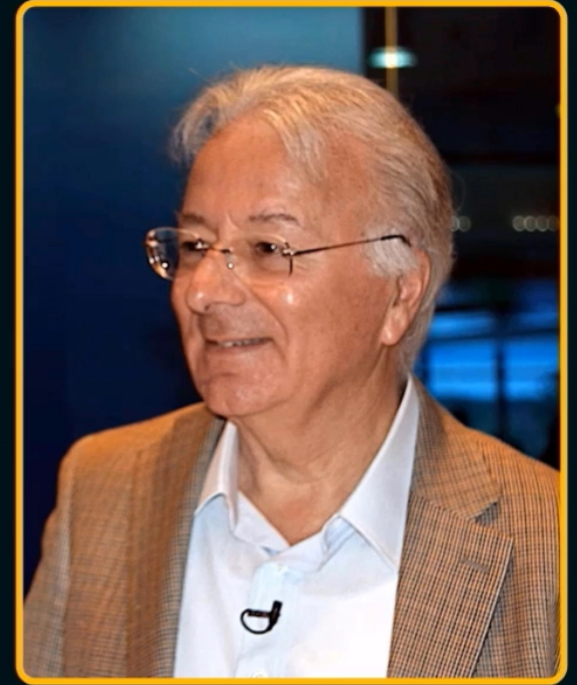
4004: 4-bit μ P

⋮



Evolution of Microprocessor

⇒ 1971: Intel 4004 4-bit μ P



Federico Faggin

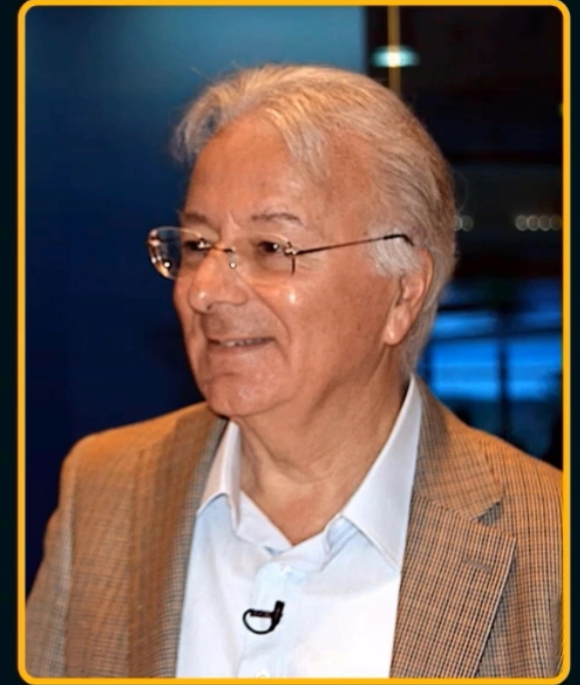


Evolution of Microprocessor

⇒ 1971: Intel 4004 4-bit μ P

00	4-bit
01	4-bit
02	4-bit
03	4-bit
...	...
79	4-bit

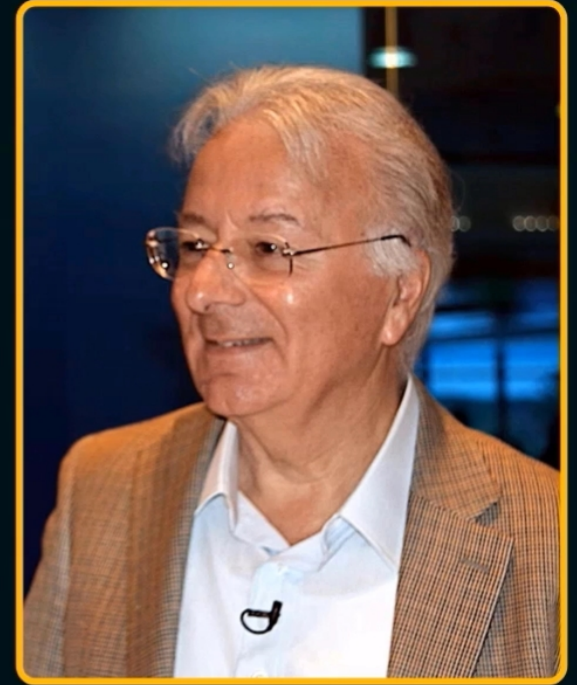
4002-RAM



Federico Faggin

Evolution of Microprocessor

- ⇒ 1971: Intel 4004 4-bit μ P
- ⇒ 1972: Intel 8008 8-bit μ P



Federico Faggin

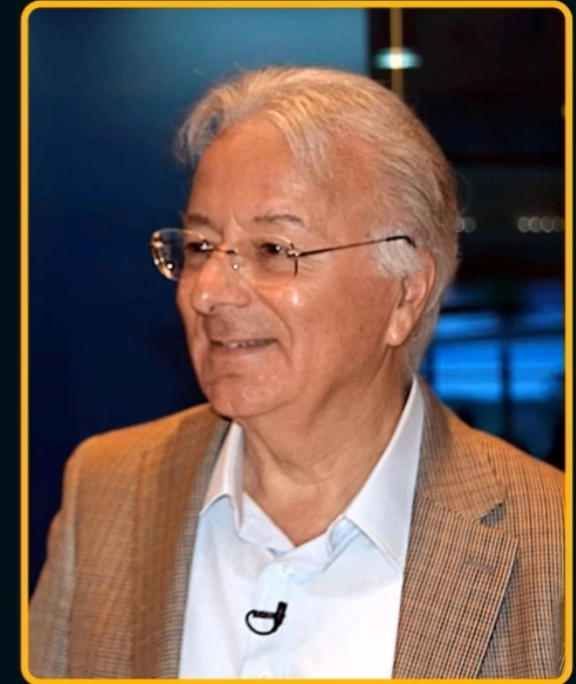
Evolution of Microprocessor

⇒ 1971: Intel 4004 4-bit μ P

⇒ 1972: Intel 8008 8-bit μ P



Mark-8 Computer Kit



Federico Faggin

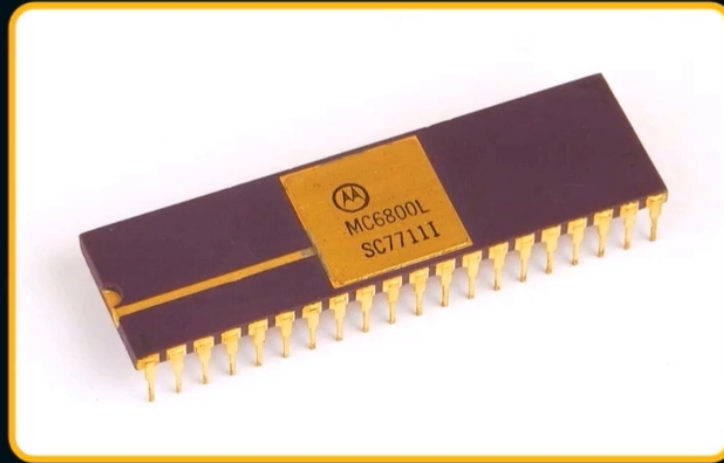


Evolution of Microprocessor

⇒ 1971: Intel 4004 4-bit μ P

⇒ 1972: Intel 8008 8-bit μ P

⇒ 1974: Intel 8080 8-bit μ P Motorola 6800

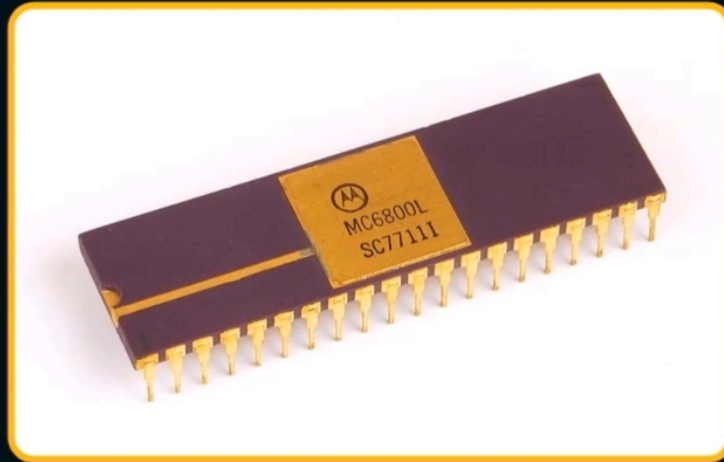


Evolution of Microprocessor

⇒ 1971: Intel 4004 4-bit μ P

⇒ 1972: Intel 8008 8-bit μ P

⇒ 1974: Intel 8080 8-bit μ P Motorola 6800, Rockwell PPS-8

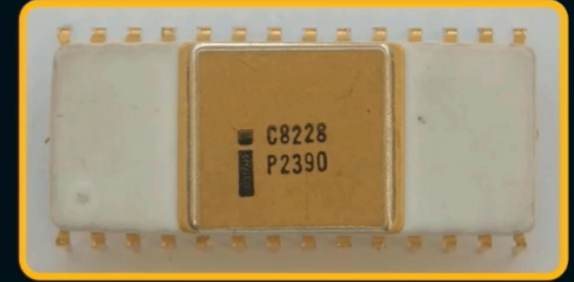


Evolution of Microprocessor

- ⇒ 1971: Intel 4004 4-bit μ P
- ⇒ 1972: Intel 8008 8-bit μ P
- ⇒ 1974: Intel 8080 8-bit μ P



8224: Clock Generator

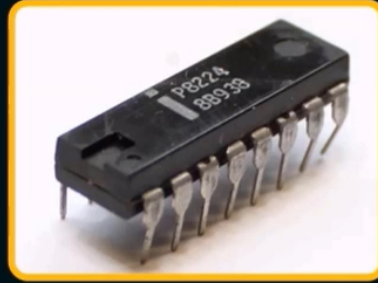


8228: System Controller

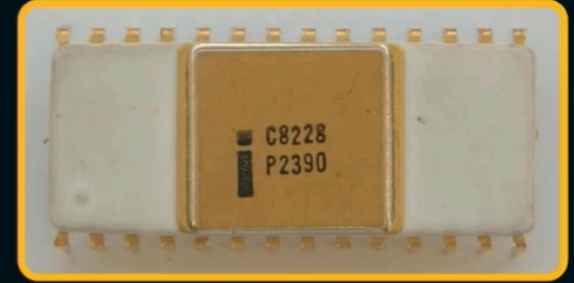


Evolution of Microprocessor

- ⇒ 1971: Intel 4004 4-bit μ P
- ⇒ 1972: Intel 8008 8-bit μ P
- ⇒ 1974: Intel 8080 8-bit μ P
- ⇒ 1977: Intel 8085 8-bit μ P



8224: Clock Generator



8228: System Controller

Evolution of Microprocessor



acadei

- ⇒ **1971**: Intel **4004** 4-bit μ P
- ⇒ **1972**: Intel **8008** 8-bit μ P
- ⇒ **1974**: Intel **8080** 8-bit μ P
- ⇒ **1977**: Intel **8085** 8-bit μ P
- ⇒ **1978**: Intel **8086** 16-bit μ P (29000 Transistors)



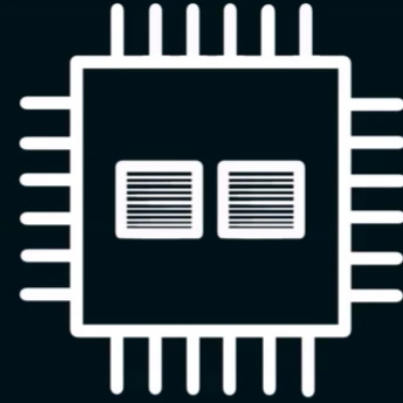
Evolution of Microprocessor

- ⇒ 1971: Intel 4004 4-bit μ P
- ⇒ 1972: Intel 8008 8-bit μ P
- ⇒ 1974: Intel 8080 8-bit μ P
- ⇒ 1977: Intel 8085 8-bit μ P
- ⇒ 1978: Intel 8086 16-bit μ P Intel 80186, Intel 80286 (134000 Transistors)
- ⇒ 1985: Intel 80386 32-bit μ P (275000 Transistors), Intel 80486 (1989)
- ⇒ 1993: Intel Pentium 1 Intel Pentium 2, Pentium 3, Pentium 4
1997 1999 2000
- ⇒ 2001: Intel Itanium 64-bit μ P



Evolution of Microprocessor

- ⇒ 1971: Intel 4004 4-bit μ P
- ⇒ 1972: Intel 8008 8-bit μ P
- ⇒ 1974: Intel 8080 8-bit μ P
- ⇒ 1977: Intel 8085 8-bit μ P
- ⇒ 1978: Intel 8086 16-bit μ P
- ⇒ 1985: Intel 80386 32-bit μ P
- ⇒ 1993: Intel Pentium 1 32-bit μ P
- ⇒ 2001: Intel Itanium 64-bit μ P



Evolution of Microprocessor

⇒ 1971: Intel 4004 4-bit μ P

⇒ 1972: Intel 8008 8-bit μ P

⇒ 1974: Intel 8080 8-bit μ P

⇒ 1977: Intel 8085 8-bit μ P

⇒ 1978: Intel 8086 16-bit μ P

⇒ 1985: Intel 80386 32-bit μ P

⇒ 1993: Intel Pentium 1 32-bit μ P

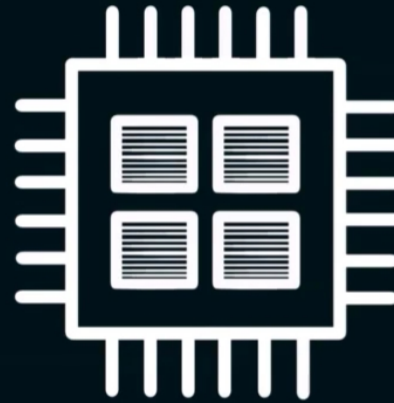
⇒ 2001: Intel Itanium 64-bit μ P

Dual
Core
 μ p



⇒ 2006: Intel Core 2 Duo, Dual Core

⇒ 2010: Intel Core i3



⇒ 2007: Intel Core 2 Quad
Intel Core i5
Intel Core i7



Quad Core μ p



Evolution of Microprocessor



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⇒ 1971: Intel 4004 4-bit μ P

⇒ 1972: Intel 8008 8-bit μ P

⇒ 1974: Intel 8080 8-bit μ P

⇒ 1977: Intel 8085 8-bit μ P

⇒ 1978: Intel 8086 16-bit μ P

⇒ 1985: Intel 80386 32-bit μ P

⇒ 1993: Intel Pentium 1 32-bit μ P

⇒ 2001: Intel Itanium 64-bit μ P

Dual
Core
 μ p

⇒ 2006: Intel Core 2 Duo, Dual Core

⇒ 2010: Intel Core i3

Quad
Core
 μ p

⇒ 2007: Intel Core 2 Quad

Intel Core i5

Intel Core i7

⇒ 2017: Intel Core i7

Intel Core i9

Octa core μ p

